



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-01 Start Here

Data Ingestion & Interfaces | exact learner route

Artifact	C3_01_START_HERE_BEGINNER_FIRST_RC1
Status	2026 - LEARNER EDITION
Teaching standard	M01 beginner-first teacher loop. Videos are visual companions; the book remains sufficient fallback.
C2B boundary	Bridge foundations may be assumed, but C3 engineering behavior is demonstrated, broken, repaired, transferred and validated.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

Course rule: preserve what arrived -> validate interpretation -> quarantine ambiguity -> publish durably -> commit state last -> prove replay safety. A green job is never sufficient evidence by itself.



What C3-01 changes

You are no longer only importing data for analysis. You are building a boundary that must prove what arrived, reject ambiguity safely, resume after failure, and support replay/backfill without losing or duplicating trusted state.

EXACT ORDER

For L01-L14: open the lesson at the exact mapped page -> follow the teacher walkthrough -> complete only matching Pxx -> freeze attempt_001 -> open matching protected review -> close it -> solve changed retry -> save evidence/mastery. After all 14 lessons, complete ShopPulse Mini-Lab, then Gate A.

NO HIDDEN INSTALLS

PyArrow first appears at L04; Requests at L05 if missing. sqlite3 is standard library; pytest already belongs to C3-00. Kafka/Debezium server are deliberately deferred to later C3 streaming modules.

C2B OVERLAP

Every C3-01 lesson is classified as deeper engineering progression. B02/B04 foundations are assumed where relevant; no lesson is copied from Bridge unchanged.

ASSESSMENT RULE

Mini-Lab and Gates use competency PASS/FAIL only. Every required competency must PASS and critical failures must equal zero. No aggregate shortcut can override a failed required competency.

PROTECTED MATERIAL

C3-022 reviews are hidden until the matching lesson attempt is saved. Gate B/C remain unseen unless explicitly assigned after diagnosis; targeted repair never auto-unlocks them.

Lesson route map

Lesson	Start page	Practice	Tool / focus
C3-01-L01	3	P01	architecture/stdlib
C3-01-L02	6	P02	csv
C3-01-L03	9	P03	json
C3-01-L04	12	P04	PyArrow first use
C3-01-L05	15	P05	Requests/local mock API
C3-01-L06	18	P06	Requests pagination
C3-01-L07	21	P07	retry policy
C3-01-L08	24	P08	sqlite3
C3-01-L09	27	P09	hash/manifests
C3-01-L10	30	P10	state tuple
C3-01-L11	33	P11	CDC semantics only
C3-01-L12	36	P12	datetime/schema policy
C3-01-L13	39	P13	filesystem state
C3-01-L14	42	P14	pytest/backfill